

Learning Equality Curriculum Recommendations

Web Mining – Course: IT4868E – Group 19

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Agenda

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- Massive amount of open educational resources (videos, exercises, documents, ...).
- Curriculum designers must manually align content to topics in a curriculum tree.
- In low-resource settings (refugee camps, rural schools), this manual process is a bottleneck.
- **Goal:** Given a curriculum topic t , automatically recommend a ranked list of relevant content items $c \in \mathcal{C}$.

Task Definition & Key Challenges

Task: Learn a scoring function

$$f : \mathcal{T} \times \mathcal{C} \rightarrow \mathbb{R}$$

such that, for each topic t , contents with higher $f(t, c)$ are more relevant.

Challenges (problem-oriented view):

- Large scale: $|\mathcal{T}| \approx 7.7 \times 10^4$, $|\mathcal{C}| \approx 1.5 \times 10^5$.
- Extreme sparsity: only $\approx 2.8 \times 10^5$ positive pairs, so

$$\pi_{\text{pos}} = \frac{N_{\text{pos}}}{|\mathcal{T}| \cdot |\mathcal{C}|} \ll 10^{-4}.$$

- Multilingual, multi-domain text (28 languages, K–12 subjects).
- Hierarchical topics, content reuse across many topics.

Dataset Overview

- `topics.csv`: topic metadata (id, title, description, language, parent_id, channel).
- `content.csv`: resource metadata (id, title, description, text, kind, language).
- `correlations.csv`: expert-labeled topic–content alignments.

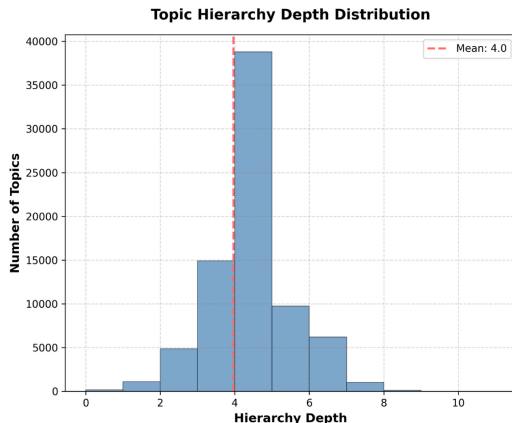
Component	Count (approx.)
Curriculum topics	76,972
Content items	154,047
Training alignments	279,919
Languages	28

EDA: Hierarchy & Sparsity

- Topics form trees of depth 3–6 by subject/grade/unit.
- Deeper topics are more specific, requiring fine-grained matching.
- Many topics have very few aligned contents (1–20).
- **Challenge:** extreme class imbalance and sparse labels.

Design implication:

- Stage 1: high-recall retrieval of a small candidate set.
- Stage 2: powerful reranker on those candidates.

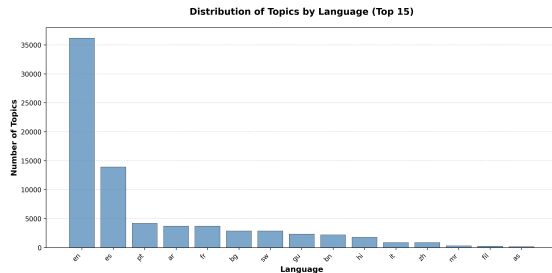


EDA: Language Distribution & Multilinguality

- English, Spanish dominate; long tail of low-resource languages.
- Most alignments are same-language; some cross-lingual matches exist.
- **Problem:** one global index risks noisy cross-lingual matches.

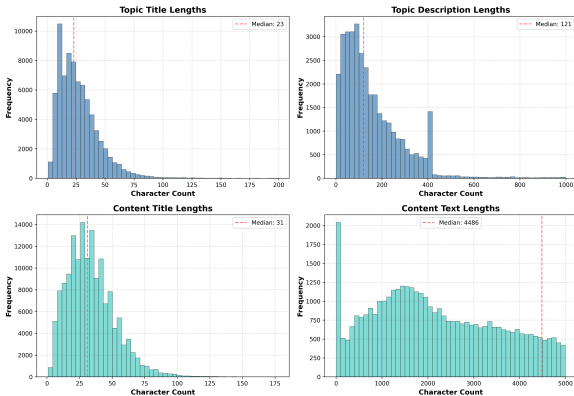
Design implication:

- Segment pipeline by language for TF-IDF retrieval.
- Use multilingual transformers (mBERT, XLM-R) for semantic matching.

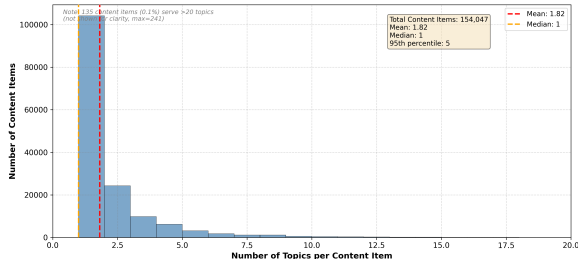


EDA: Text Length & Content Reuse (1/2)

Text Length Distributions



Distribution of Topic Counts per Content Item



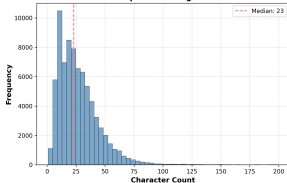
Key observations:

- Topic titles are very short; descriptions vary a lot in length.
- Some content items serve many topics (shared foundational resources).

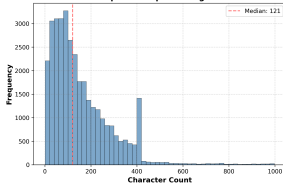
EDA: Text Length & Content Reuse (2/2)

Text Length Distributions

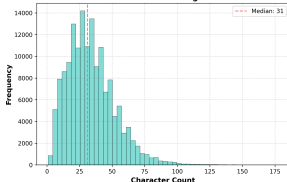
Topic Title Lengths



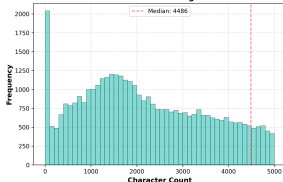
Topic Description Lengths



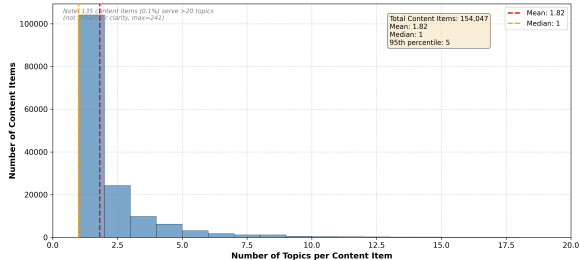
Content Title Lengths



Content Text Lengths



Distribution of Topic Counts per Content Item



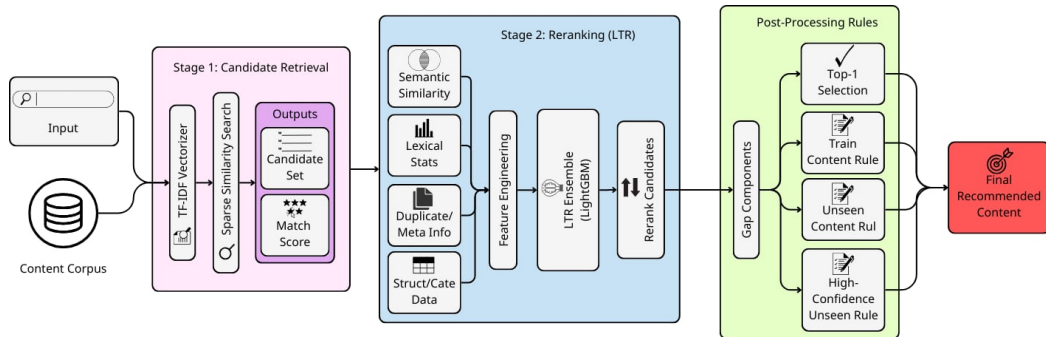
Design implication:

- Build composite topic text: ancestor titles + title + description.
- Engineer features based on content reuse and duplicate patterns.

From EDA to architecture:

- Scalability $\Rightarrow$ **Two-stage IR pipeline**: fast retrieval + heavy reranking.
- Multilingual & noisy text $\Rightarrow$ **Char 4-gram TF-IDF** + multilingual embeddings.
- Hierarchy & short topics $\Rightarrow$ **Composite topic text** with ancestor titles.
- Sparse labels & content reuse $\Rightarrow$ **Duplicate / second-degree features**.
- Evaluation focus on recall (F_2) $\Rightarrow$ Stage 1 tuned for high recall, Stage 2 + post-processing handle precision.

Overall Two-Stage Pipeline



- Stage 1: TF-IDF Vector Space Model retrieves top- N candidates per topic.
- Stage 2: Feature-based LightGBM reranker with multilingual embeddings and structural signals.
- Post-processing: score-gap rules to trade off precision vs recall.

Preprocessing & Text Representation

- Normalize text: lowercase, Unicode normalization, clean markup.
- Preserve punctuation and math symbols (important for STEM content).
- Composite topic text:

$$\text{text}_{\text{topic}} = \text{ancestor titles} \parallel \text{title} \parallel \text{description}.$$

- Composite content text:

$$\text{text}_{\text{content}} = \text{title} \parallel \text{description} / \text{truncated text}_{\leq L}.$$

- Language-aware segmentation: build separate indices per language.

TF-IDF Char 4-gram VSM

Why char 4-grams?

- Robust across languages, spelling variants, tokenization.
- Captures subword patterns and mathematical notation.

TF-IDF weight:

$$\text{tfidf}(w, d) = \text{tf}(w, d) \cdot \log \left(\frac{N}{\text{df}(w)} \right),$$

where w is a 4-gram, d is topic or content text.

Similarity:

$$\text{sim}_{\text{tfidf}}(t, c) = \cos(\mathbf{x}_t, \mathbf{x}_c) = \frac{\mathbf{x}_t^\top \mathbf{x}_c}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_c\|_2}.$$

Efficient Retrieval & Hyperparameters

- Sparse matrices: T (topics) and C (contents), all rows L2-normalized.
- Compute top- N cosine similarities using `awesome_cossim_topn`.
- For each topic t :

$$\mathcal{C}_N(t) = \{c \mid c \text{ is among top-}N \text{ by } \text{sim}_{\text{tfidf}}(t, c) \text{ and } \text{sim}_{\text{tfidf}}(t, c) \geq \tau\}.$$

- **Key hyperparameters:**
 - $N = 20$ candidates per topic (default).
 - Threshold $\tau = 0.01$.
- These values are chosen so that Stage 1 achieves very high Recall@20/50 while keeping the candidate set small for Stage 2.

Feature Groups (1): Semantic vs Lexical Similarity

Semantic similarity features (“Semantic” group in ablations):

- Multilingual encoders:
 - bert-base-multilingual-uncased (mBERT).
 - xlm-roberta-base.
 - xlm-roberta-large.
- Weighted concatenated embedding for any text x :

$$e(x) = [0.3 e_{\text{mBERT}}(x), 0.3 e_{\text{XLM-R-base}}(x), 0.4 e_{\text{XLM-R-large}}(x)].$$

- Distance-based features for each pair (t, c) :

$$\text{vec_dist}(t, c) = 1 - \cos(e(t), e(c)),$$

plus topic-level aggregates such as `vec_dist_min`, `vec_dist_max`, `vec_dist_mean`.

Lexical similarity features (“Lexical” group in ablations):

- `match_score(t,c)`: TF-IDF cosine similarity from Stage 1.
- Aggregates such as `match_score_max`, `match_score_min`, `match_score_mean`.
- Role: complement semantic embeddings when topics are very short or content is heavy in formulas and symbols.

Feature Groups (2): Duplicate/Graph vs Structural/Length

Duplicate & graph-based signals (“Duplicate-Graph” group):

- Reuse counts: `dup_count`, `tdup_count`, `pdup_count`.
- `second_degree`: content c is already used by a neighbor topic (within the curriculum tree) of t .
- Training-set signals: `content_is_train`, `content_max_train_score`, `topic_max_train_score`.
- Idea: encode the EDA insight that heavily reused content in training tends to be high-quality alignment anchors.

Structural & length features (“Structural-Length” group):

- Curriculum structure: `same_chapter`, `starts_same`, topic depth in the tree.
- Text length features: `len_topic_title`, `len_topic_description`, `len_content_title`, `len_content_text`, ...
- Categorical metadata: `language`, `channel`, `kind` (one-hot encoded).
- Role: provide soft priors that penalize unusual matches (e.g., very short topic vs extremely long content, or mismatched channel).

Learning-to-Rank with LightGBM

Labels:

$$y_{t,c} = \begin{cases} 1 & \text{if } c \text{ is aligned with } t, \\ 0 & \text{otherwise (candidate from Stage 1).} \end{cases}$$

Model:

- Gradient-boosted trees (LightGBM) with binary objective.
- Loss:

$$\mathcal{L}(\theta) = - \sum_{(t,c)} [y_{t,c} \log \sigma(s_{t,c}) + (1 - y_{t,c}) \log(1 - \sigma(s_{t,c}))],$$

where $s_{t,c}$ is the raw score and σ is the sigmoid.

Monotonic constraint:

- Enforce that larger `vec_dist` $\Rightarrow$ lower predicted relevance.
- Encodes prior knowledge from semantic embeddings directly into the tree model.

Key Hyperparameters & CV Strategy

LightGBM hyperparameters:

- `learning_rate`: 0.01–0.02.
- `num_boost_round`: 1600–3200.
- `num_leaves`: 31, `min_data_in_leaf`: 1024.
- `feature_fraction` = 0.8, `subsample` = 0.5.
- `is_unbalance` = True to handle label imbalance.

Why these choices?

- Low learning rate + many trees: stable training on a large, noisy dataset.
- Strong regularization (large `min_data_in_leaf`, subsampling) to avoid overfitting.
- GroupKFold by `topic_id`: no leakage of the same topic text across train/validation.

Embedding-level ensemble

- Combine mBERT, XLM-R base, XLM-R large.
- Motivation: different pretraining corpora capture complementary semantics across languages and domains.

LTR-level ensemble

- 8 LightGBM models:
 - 4 folds with the full feature set.
 - 4 folds with a reduced feature set (no aggregate features).
- Final score:

$$s_{\text{ensemble}}(t, c) = \frac{1}{8} \sum_{m=1}^8 g_m(f(t, c)).$$

- Problem-oriented view: averaging across folds and feature subsets reduces variance and stabilizes ranking, especially for low-resource languages/topics.

Post-processing Rules

Score gap:

$$\text{gap}(c) = \max_{c' \in \mathcal{C}_N(t)} s(t, c') - s(t, c).$$

Rules per topic t :

- Always include the top-1 candidate.
- For `content_is_train`:

$$\text{gap}(c) < 0.25, \quad s(t, c) > 0.05.$$

- For unseen content:
 - Include if it is the best match for that content among all topics and the content-centric distance is small.
 - Or if $\text{gap}(c) < 0.05$ (very high confidence).

Idea: keep recall high but avoid over-assigning weak matches, especially for new content.

Evaluation Metrics

Official metric: micro- F_2

$$F_2 = \frac{(1 + 2^2)PR}{2^2P + R} = \frac{5PR}{4P + R}.$$

- Recall-focused: missing relevant content is penalized more heavily than recommending extra items.

Additional IR metrics & their roles

Metric	Symbol	Used for
Micro- F_2 score	F_2	Main competition metric; model selection and leaderboard comparison.
Recall@K	Recall@K	Diagnose Stage 1: does the candidate pool cover relevant content?
Precision@K	Precision@K	Compare post-processing strategies and precision/recall trade-offs.
Mean Reciprocal Rank	MRR	Position of the first relevant item; effectiveness of Stage 2 reranking.
NDCG@10	NDCG@10	Quality of the top-10 ranking; approximates real user behavior.
ROC-AUC	AUC	Training-time metric for LightGBM (ability to separate positives/negatives).

Cross-validation: 4-fold GroupKFold by `topic_id` to simulate unseen topics at test time.

Overall Performance: Validation & Test Set

Table 1: Offline 4-fold GroupKFold validation metrics.

Config	Backbone	Emb Dim	#LGB	F_2 (CV)	AUC	MRR	NDCG@10	Recall@50
Single-mBERT	mBERT-base	768	4	0.595	0.86	0.79	0.73	0.92
Single-XLMR-Base	XLM-R base	768	4	0.601	0.87	0.81	0.75	0.93
Single-XLMR-Large	XLM-R large	1024	4	0.612	0.89	0.84	0.80	0.95
Final Ensemble	3-Backbone	2560	8	0.617	0.89	0.86	0.81	0.96

Table 2: Test-set performance (Public vs Private).

Config	Backbone	#LGB	F_2 (Pub)	F_2 (Priv)
Single-mBERT	mBERT-base	4	0.593	0.591
Single-XLMR-Base	XLM-R base	4	0.599	0.597
Single-XLMR-Large	XLM-R large	4	0.608	0.606
Final Ensemble	3-Backbone	8	0.617	0.617

Ablation: Stage 1 Retrieval Size & Backbone Capacity

1. Stage 1 candidate pool size

Goal: Choose N to balance recall and inference cost.

Config	#Candidates	F_2 (CV)	Recall@50
S1-Top10	10	0.603	0.93
S1-Top20 (Default)	20	0.617	0.96
S1-Top50	50	0.617	0.96

2. Backbone capacity

Goal: Is a base backbone sufficient, or do we need XLM-R large with two-stage training?

Backbone	Dim	F_2 (CV)	AUC	NDCG@10
Single-mBERT	768	0.595	0.86	0.73
Single-XLMR-Base	768	0.601	0.87	0.75
Single-XLMR-Large	1024	0.612	0.89	0.80

Ablation: Embedding Weights & Feature Groups

3. Embedding weight configurations

Goal: Understand the impact of $(w_{\text{mBERT}}, w_{\text{Base}}, w_{\text{Large}})$.

Config ($w_{\text{mBERT}}, w_{\text{Base}}, w_{\text{Large}}$)	F_2 (CV)	AUC	MRR	NDCG@10
No-Large (0.5, 0.5, 0.0)	0.608	0.88	0.84	0.79
Equal-Weights (0.33, 0.33, 0.33)	0.613	0.89	0.85	0.80
Default (0.3, 0.3, 0.4)	0.617	0.89	0.86	0.81
Large-Heavy (0.2, 0.3, 0.5)	0.616	0.89	0.86	0.81

4. Feature group importance

Goal: Identify which feature groups are critical for Stage 2 performance.

Config	F_2 (CV)	AUC	MRR	NDCG@10	Recall@50
Full-Features	0.617	0.89	0.86	0.81	0.96
No-Semantic	0.588	0.87	0.81	0.79	0.96
No-Lexical	0.606	0.88	0.85	0.80	0.95
No-Duplicate-Graph	0.612	0.89	0.85	0.81	0.96
No-Structural-Length	0.613	0.89	0.86	0.81	0.96

Ablation: Ensemble Depth & Post-processing

5. Ensemble depth & diversity

Goal: How many LGBM models and backbones are enough?

Config	Backbones	#LGBM	F_2 (CV)	AUC	NDCG@10
Single-Large-1LGBM	XLM-R large	1	0.608	0.88	0.79
Single-Large-4LGBM	XLM-R large	4	0.612	0.89	0.80
TwoBackbone-4LGBM	mBERT + XLM-R Lg	4	0.615	0.89	0.81
ThreeBackbone-8LGBM	mBERT + XLM-R Base + Lg	8	0.617	0.89	0.81

6. Post-processing strategy

Goal: Effect of rules on precision/recall and F_2 .

Config	F_2 (CV)	Precision	Recall	MRR	NDCG@10
NoPost (Top-10)	0.607	0.52	0.79	0.84	0.80
Post-Top1Only	0.611	0.54	0.80	0.85	0.81
Post-Default	0.617	0.50	0.84	0.86	0.81
Post-Strict	0.610	0.55	0.76	0.85	0.81

Takeaway:

- Stage 1 solves the *retrieval sufficiency* problem under computational constraints.
- Stage 2 + ensemble solves the *ranking accuracy* problem using semantic, lexical, duplicate-graph, and structural-length signals suggested by EDA.

Summary of Contributions:

- Developed a scalable pipeline combining high-recall lexical retrieval (TF-IDF) with precise semantic reranking (LightGBM + Transformers).
- Achieved strong performance on the micro-averaged F_2 metric through careful feature engineering and multilingual ensembling.
- Ensured inference efficiency using offline embedding precomputation and limited candidate pools.

Team Members & Contributions

No.	Full Name	Student ID	Contribution	Main Tasks
1	Nguyen Hai Dang	20225478	20%	mBERT, LightGBM modeling
2	Nguyen Vo Ngoc Khue	20225504	20%	XLNet Base, feature engineering
3	Nguyen Hoang Quan	20225519	20%	XLNet Large, two-stage training
4	Nguyen Tien Dat	20225481	20%	TF-IDF retrieval, ensemble pipeline
5	Phan Minh Hoa	20225495	20%	GUI, documentation

Thank You for Your Attention!

Q & A

Group 19 – Web Mining (IT4868E)